

ESPNOWMIDIBridge Firmware

ESPNO'WMIDIBridge: ESP32-C6 Embedded Swift Firmware

This is the firmware component for the ESP-NO'W to ALSA MIDI bridge. Runs great on an ESP32-C6 microcontroller.

Overview

The ESPNO'WMIDIBridge receives ESP-NO'W frames from wireless sensor/controller nodes, translates them to MIDI events according to a configurable mapping table, and forwards the events to a host running ALSAMIDISource over USB CDC-ACM. ALSAMIDISource then injects these MIDI events into the ALSA sequencer.

Architecture

The firmware is organized into three loosely coupled components:

- > Radio: ESP-NO'W initialization, peer management, frame reception
- > Translator: Mapping ESP-NO'W payloads to MIDI events
- > SerialTransport: COBS encoding and USB CDC-ACM transmission

Configuration

Compile-Time Configuration

Edit `main/Kconfig.projbuild` or use `idf.py menuconfig`:

- > Room ID: Bridge identifier for peer discovery (default: 0x01)
- > WiFi Channel: ESP-NO'W channel (default: 1)
- > Advertisement Interval: How often bridge announces itself (default: 5000 ms)
- > Peer Timeout: When to remove inactive peers (default: 30000 ms)
- > Max Peers: Peer table size (default: 10)

Mapping Rules

Edit `main/Translator.swift` in the `MappingConfig.rules` array to define how ESP-NOW payloads map to MIDI events.

Each rule specifies:

- > Payload byte(s) to match (offset, mask, expected value)
- > Optional sender MAC filter
- > Target MIDI channel and event type
- > How to derive MIDI data bytes (from payload, MAC, or fixed literals)
- > Optional SysEx payload mapping

Example mapping from payload byte 0 == 0x10 to Note On:

```
MappingRule(  
    matchOffset: 0,  
    matchMask: 0xFF,  
    matchValue: 0x10,  
    senderMacFilter: nil,  
    midiChannel: 0,  
    midiEventType: .noteOn,  
    data1Source: .payloadByte(1),      // Note number  
    data2Source: .payloadByte(2),      // Velocity  
    sysExPayloadStart: nil,  
    sysExPayloadLen: nil  
)
```

Wire Protocol

The bridge communicates with the host using COBS-framed MIDI event frames over USB CDC-ACM at 921600 baud.

Frame Structure

[COBS-encoded MIDIEventFrame] 0x00

Each frame contains:

- > Sender MAC address (6 bytes)
- > MIDI channel (1 byte)
- > MIDI event type (1 byte)
- > Data byte 1 (1 byte)
- > Data byte 2 (1 byte)
- > SysEx length and data (variable)

Testing

The firmware can be tested using the [Tildagon MIDI Controller](#).

See the ALSAMIDISource package for host-side usage.